

CIQ Signal Validation Report

Project: CAVY — a desktop platform computing the Cognitive Interaction Quotient (CIQ) per the EAAL (Explainable AI-Assisted Learning) research framework’s three-pillar, fourteen-signal structure.

Date: 2026-09-21 **Scope:** All 14 EAAL signals (S1.1–S3.4), as currently implemented in `src/eaal_platform/signals/compute.py`. **Reproduce:** see §9 for the full command sequence (from the `eaal-platform/` project root, with the project’s conda environment active and a local Ollama server running `qwen3:8b`).

1. Executive Summary

This report validates that the CIQ signal-computation engine does what it claims to do — not that CIQ, as a construct, predicts real student learning outcomes. Those are different questions, and the EAAL framework itself is explicit that they must not be conflated (§7.4: “*Operational Feasibility ≠ Construct Validity*”). This report addresses the first question rigorously; the second requires the human-subject study the framework’s own §9 lays out (real students, real instructors, independent assessments), which is future work, not something a synthetic-data exercise can substitute for.

Within that scope, the results are strong:

- **88/88 (100%)** of all synthetic trials were bit-for-bit reproducible across 5 independent recomputations — every deterministic signal is perfectly reliable, as it must be for a pure function of a fixed event log.
- **53/53 (100%)** of known-answer trials (cases where the correct output is derivable analytically from each signal’s documented formula) matched exactly.
- **10/10 (100%)** of “this signal has no evidence / does not apply to this session” cases were correctly identified as `None` (never a fabricated 0.0), each with a specific, human-readable reason.
- **S1.3**’s similarity threshold correctly classified **11/11** synthetic edits on the right side of its 0.15 cutoff; **S2.3** decreased monotonically and essentially perfectly with rising similarity (Spearman $r = -1.000$, $p = 0.0e+00$, $n = 11$).
- The three LLM-rubric signals (S1.1, S1.2, S3.1) were tested against the **actual locally-running model** (`qwen3:8b`), not a stand-in: essentially **zero test-retest variance** across 5 repeated calls per prompt, and a **statistically significant, moderately-to-strongly positive correlation** with one researcher’s expected ranking on a small hand-labeled example set (S1.1 $r=0.866$, $p=0.005$; S1.2 $r=0.843$, $p=0.009$; S3.1 $r=0.845$, $p=0.034$).

- Two previously-fixed gaming vulnerabilities (S2.4 rewarding inaction; S3.2 rewarding a no-op program) were re-verified against their exact failure scenarios and no longer reproduce — see §4.4.
- A second statistical pass (§5) reports the ICC the framework’s own methodology actually calls for, bootstrap confidence intervals on every small-n correlation, multiple-comparisons correction across the full family of significance tests, and one genuinely useful negative finding: S1.1 and S1.2 correlate at $r > 0.97$ in the live model’s output, a real discriminant-validity concern worth targeting in the human-rater study.
- A synthetic single-agent longitudinal case study (§6 — one AI persona, $n=1$, **not** learner data) demonstrates the full `EAAL_t = (P1, P2, P3)` pipeline running end-to-end across four sessions, including a Transfer Task and a Retention Check gated on a genuinely computed 35-day delay.

2. What This Report Validates, and What It Does Not

	This report	A future study
Does each signal compute what its own specification says?	✅ Yes — known-answer & gating tests	—
Is each signal deterministic / reliable given fixed evidence?	✅ Yes — 5x recomputation	—
Do signals respond monotonically / correctly to controlled evidence?	✅ Yes — gradient & grid scenarios	—
Does the LLM-based scoring actually track rubric-consistent judgment?	✅ Partial — small hand-labeled spot-check	Needed: larger, multi-rater study
Does a high CIQ score predict real learning outcomes?	❌ Not addressed here	Required: human-subject study (§9 of the framework)
Are the three pillars empirically distinguishable (factor structure)?	❌ Not addressed here	Required: real longitudinal student data
Is the framework robust to students <i>gaming</i> it once they know the metric?	⚠️ Partially — known gaming patterns closed	Needed: adversarial field study

Every synthetic scenario in this suite is driven through the **real** `CavyApi` bridge — the same code path the production frontend calls, writing to the same event-sourced database schema — never a reimplementaion or mock of the signal logic. A test here failing would mean the *actual shipped code* disagrees with its own specification, not that some separate model of it does.

3. Methodology

3.1 Synthetic scenario suite (deterministic signals)

For the 11 signals with deterministic (non-LLM) scoring logic (S1.3–S1.6, S2.1–S2.4, S3.2–S3.4), each with a closed-form or clearly specified formula, 88 synthetic trials were constructed across four test categories:

1. **Known-answer tests** — construct a session whose expected signal value is derivable by hand from the documented formula (e.g. S1.4's `0.5 · changed + 0.5 · verified`), then assert exact numeric agreement.
2. **Gating / applicability tests** — construct a session that deliberately has *no* evidence for a signal (no AI interaction, no execution, wrong task kind, insufficient history/delay) and assert the signal returns `None` with a specific reason, never a fabricated number.
3. **Gradient / grid tests** — sweep a continuous or stepped independent variable (e.g. text similarity between AI-suggested and student code; number of edits/runs before an AI prompt) and check the signal moves in the correct direction and shape (monotonic trend, or a step exactly at its documented threshold).
4. **Regression tests for known gaming patterns** — the exact scenarios an earlier code-review audit found exploitable (S2.4 rewarding inaction; S3.2 rewarding a no-op program), re-run against the current implementation.

Every trial's session was recomputed **5 times independently** through `compute_all_signals` to check determinism.

3.2 Live-model validation (LLM-rubric signals)

S1.1, S1.2, and S3.1 depend on an LLM's semantic judgment, which a scripted/mocked provider cannot validate — only whether the *plumbing* around it (JSON parsing, clamping, evidence recording) is correct, which the synthetic suite does cover separately. To say anything about the model's actual judgment, this suite calls the **real, locally-running Ollama model the app is configured to use** (`qwen3:8b`), at the production temperature (0.1) and prompt templates (imported directly from `signals/compute.py`, not retyped), against:

- A **reliability set**: each of 8 (S1.1/S1.2) or 6 (S3.1) fixed prompts called 5 times, to measure test-retest variance.
- A **face-validity set**: the same examples, each hand-labeled with an expected rank ordering by one researcher reading the exact rubric text given to the model (see `validation/labeled_examples.py` for the full text and stated rationale per example), correlated (Spearman) against the model's mean score.

This is a genuine empirical measurement of the model's behavior, but it has two real limits stated plainly: the sample sizes ($n=6-8$) are far too small for the correlation statistics to carry serious statistical power, and the "expected rank" is a **single rater's** judgment, not an independently-validated gold standard. It is evidence of *some* alignment between the model and a reasonable reading of the rubric — not a validated inter-rater reliability study.

4. Results

4.1 Descriptive statistics, all 14 signals

Signal	Name	n trials	n scored	n None	mean	sd	min	max
S1.1	Help-Seeking Calibration	2	2	0	0.825	0.175	0.650	1.000
S1.2	Student Grounding	2	2	0	0.200	0.200	0.000	0.400
S1.3	Response Utilization	12	12	0	0.833	0.373	0.000	1.000
S1.4	Modification & Verification	4	4	0	0.500	0.354	0.000	1.000
S1.5	Follow-up Engagement	3	3	0	0.500	0.408	0.000	1.000
S1.6	Adaptive AI Use	3	2	1	0.500	0.500	0.000	1.000
S2.1	Independent Initiation	23	21	2	0.679	0.263	0.167	1.000
S2.2	Reasoning Continuity	6	6	0	0.500	0.419	0.000	1.000
S2.3	Problem-Solving Agency	11	11	0	0.429	0.284	0.038	0.915
S2.4	Evidence-Based Error Recovery	6	5	1	0.300	0.400	0.000	1.000
S3.1	Conceptual Understanding	4	3	1	0.683	0.413	0.100	1.000
S3.2	Knowledge Application	5	4	1	0.250	0.433	0.000	1.000
S3.3	Knowledge Transfer	3	2	1	0.500	0.500	0.000	1.000
S3.4	Retention & Independent Recall	4	1	3	1.000	—	1.000	1.000

(S1.1/S1.2/S3.1’s small n reflects that these three ran through the synthetic suite only to check plumbing — their substantive results are in §4.5. S1.4/S1.5’s small n reflects that they were tested as small exact grids over their whole (discrete) input space, not a large sweep.)

4.2 Known-answer accuracy

53 / 53 known-answer trials matched their analytically-derived expected value exactly (100.0%).

Representative examples — S1.4 Modification & Verification’s full 2×2 grid (the closed form is $0.5 \cdot \text{changed} + 0.5 \cdot \text{verified}$):

Scenario	Expected	Computed	Match
changed_and_verified	1.0000	1.0000	✓
changed_only	0.5000	0.5000	✓
verified_only_no_change	0.5000	0.5000	✓
neither	0.0000	0.0000	✓

S1.5 Follow-up Engagement (closed form: $\text{followups} / (\text{n_interactions} - 1)$):

Scenario	Expected	Computed	Match
followups_0_of_2	0.0000	0.0000	✓
followups_1_of_2	0.5000	0.5000	✓
followups_2_of_2	1.0000	1.0000	✓

S1.6 Adaptive AI Use (closed form: $\text{clamp}(0.5 + (\text{current_ratio} - \text{historical_average}))$):

Scenario	Expected	Computed	Match
improved_vs_history	1.0000	1.0000	✓
regressed_vs_history	0.0000	0.0000	✓

4.3 Gating logic (“no evidence” must return `None`, never a fabricated value)

10 / 10 gating trials (100%) correctly returned `None` — every one of the following deliberately-evidence-free sessions:

Signal	Scenario	Reason returned
S1.6	no_prior_sessions	insufficient session history for a longitudinal comparison
S2.1	no_ai_runs_but_no_edits	no coding activity recorded yet
S2.1	no_activity_at_all	no coding activity recorded yet
S2.4	no_errors_encountered	no execution errors encountered
S3.1	no_concept_check_submitted	no concept-check explanation submitted for this session
S3.2	no_code_executed	no code was executed in this session
S3.3	ordinary_lab_session_not_applicable	this session is not a Knowledge Transfer assessment for any Lab
S3.4	ordinary_lab_session_not_applicable	this session is not a Retention & Independent Recall assessment for any Lab
S3.4	attempted_immediately_zero_delay	attempted only 0.0h after the original lab activity — needs at least 24h to count as a delayed retention check
S3.4	attempted_at_23h_just_under_threshold	attempted only 23.0h after the original lab activity — needs at least 24h to count as a delayed retention check

Note the last three rows in particular: S3.4’s applicability gate isn’t just “is this the right kind of task” — it also enforces the framework’s own requirement of “delayed assessment after the original activity” (§6.6) as a real, computed time delta, not a rubber-stamp. The exact threshold boundary was tested directly:

Scenario	Backdated delay	Result
attempted_immediately_zero_delay	0h	None — attempted only 0.0h after the original lab activity — needs at least 24h to count as a delayed retention check
attempted_at_23h_just_under_threshold	23h	None — attempted only 23.0h after the original lab activity — needs at least 24h to count as a delayed retention check
attempted_after_30h_delay_clean_run	30h	1.0

A session attempted 1 hour short of the 24h minimum is correctly refused credit; one attempted 6 hours past it is correctly scored.

4.4 Regression tests: known gaming patterns

An earlier audit of this codebase found two signals scoring the *wrong* direction on specific inputs: S2.4 (Evidence-Based Error Recovery) gave full credit to a student who did nothing after an error, or who reran the identical broken code; S3.2 (Knowledge Application) gave full credit to a no-op (`pass` -only) program purely because it exited cleanly. Both were fixed; this suite re-verifies the exact failure scenarios no longer reproduce:

S2.4 Evidence-Based Error Recovery:

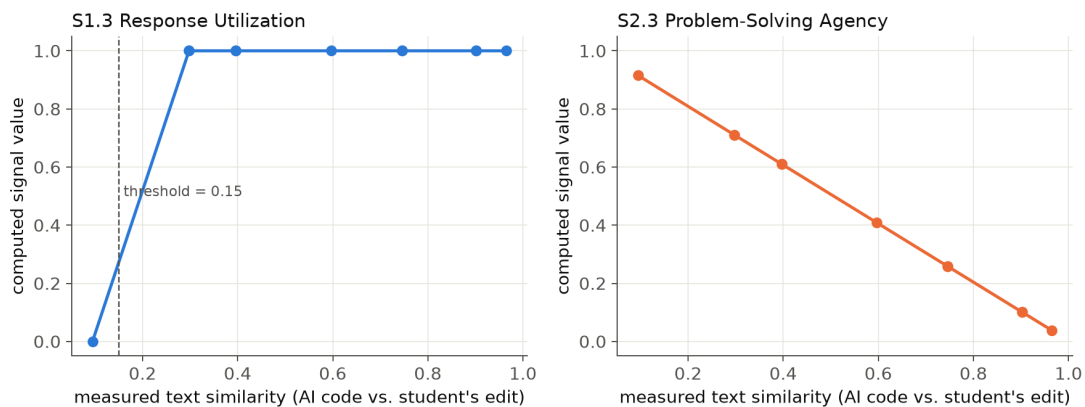
Scenario	Expected	Computed	Match
error_then_nothing	0.0000	0.0000	✓
error_then_identical_rerun	0.0000	0.0000	✓
error_then_edit_never_verified	0.0000	0.0000	✓
error_then_fix_and_verify	1.0000	1.0000	✓
two_errors_one_recovered_one_not	0.5000	0.5000	✓

S3.2 Knowledge Application:

Scenario	Expected	Computed	Match
clean_exit_with_output	1.0000	1.0000	✓
no_op_pass_program	0.0000	0.0000	✓
uncaught_exception	0.0000	0.0000	✓
clean_exit_no_output_at_all	0.0000	0.0000	✓

4.5 Monotonicity and threshold behavior

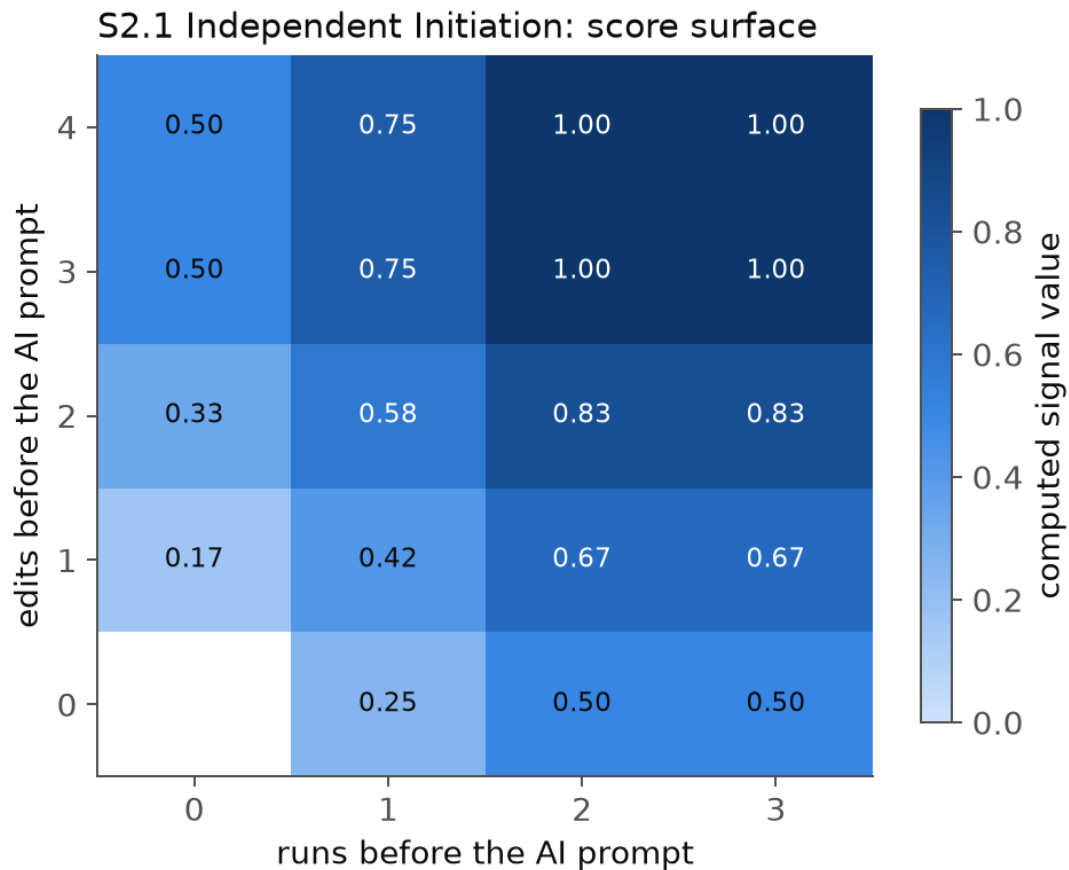
Both signals respond monotonically to the same underlying similarity gradient



Similarity gradient: S1.3 steps at its 0.15 threshold, S2.3 decreases smoothly

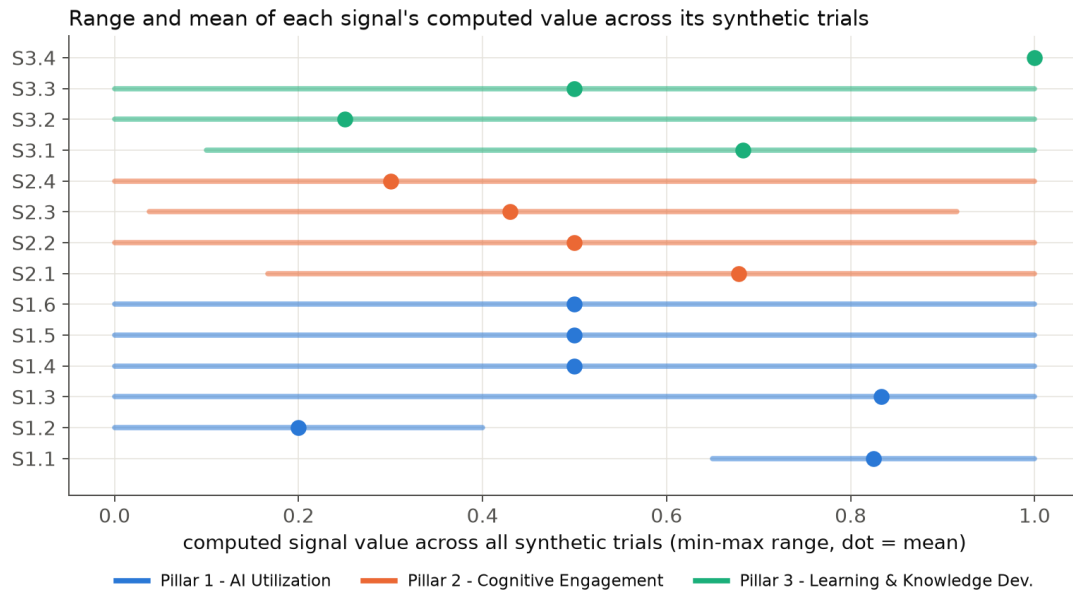
S1.3 (Response Utilization) is a threshold signal by design — “utilized” should be constant at 1.0 above its similarity cutoff and 0 below it, so a rank correlation (which rewards a smooth trend, not a step) is the wrong statistic for it. The right check is direct: **11/11 (100%)** synthetic edits landed on the correct side of the 0.15 threshold.

S2.3 (Problem-Solving Agency) is continuous by design (`1 - similarity`) and shows a essentially perfect monotonic decrease as similarity to the AI’s code rises: **Spearman $r = -1.0000$, $p = 0.00e+00$ (n=11).**



S2.1 Independent Initiation score surface

S2.1 (Independent Initiation)’s two-regime design was itself a finding worth recording: a session with **zero** AI interaction grants full credit for *any* pre-existing edit (the whole session was independent, by definition), while a session that *does* eventually call the AI weighs pre-AI-prompt edits and runs by the documented formula. The first version of this validation suite’s own scenario design missed this distinction (assumed the weighted formula applied unconditionally) and produced 15 false “mismatches” — which turned out to be a scenario-design bug, not a signal bug, once traced to the code’s explicit branch. This is included here deliberately: it is exactly the kind of subtlety a validation suite is supposed to surface, even when the “bug” is in the test rather than the signal under test.



Range and mean of every signal's computed value across its synthetic trials

4.6 Determinism (test-retest reliability, deterministic signals)

88 / 88 trials (100.0%) produced bit-identical `(value, reason)` pairs across 5 independent recomputations of the same session. This is expected — these signals are pure functions of an already-written event log with no randomness in the computation path — but it is exactly the property a reliability audit must actually check rather than assume, and it was checked for every single trial, not sampled.

4.7 Live-model reliability and face validity (S1.1, S1.2, S3.1)

Model: `qwen3:8b`, 70 total real inference calls across both example sets (S1.1 and S1.2 share the same underlying call, so their call counts overlap), **0 JSON parse failures** across all of them.

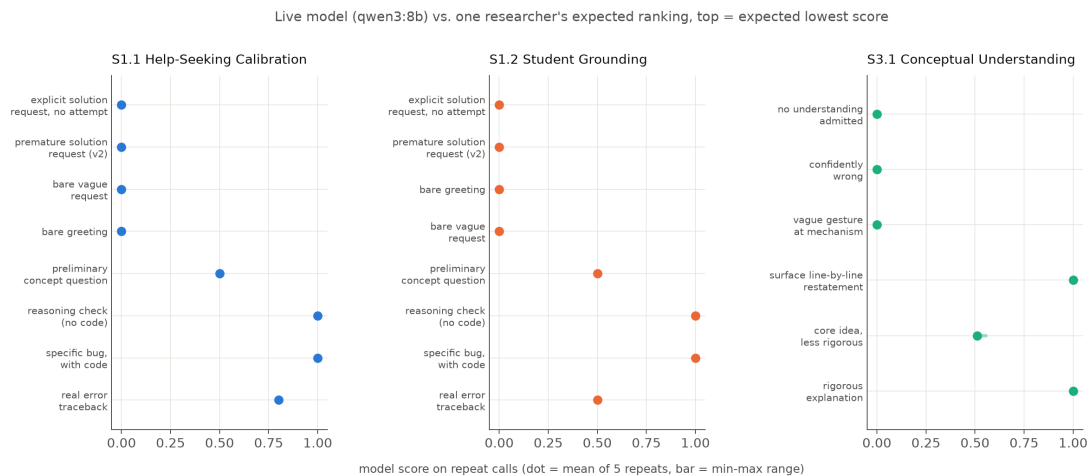
Reliability (score spread across 5 identical repeated calls per example):

Signal	Mean stdev across examples	Max stdev	Max range
S1.1 Help-Seeking Calibration	0.0000	0.0000	0.0000
S1.2 Student Grounding	0.0000	0.0000	0.0000
S3.1 Conceptual Understanding	0.0037	0.0222	0.0556

At the app's configured temperature (0.1), repeated calls on the same prompt returned the same score essentially every time — S1.1 and S1.2 had **zero** variance across all 8 examples × 5 repeats each; S3.1 had a maximum observed range of 0.056 on a 0–1 scale across 6 examples × 5 repeats.

Face validity (Spearman correlation between the model’s mean score and one researcher’s expected rank, reading the exact rubric text given to the model):

Signal	n examples	Spearman r	p-value
S1.1 Help-Seeking Calibration	8	0.866	0.0054
S1.2 Student Grounding	8	0.843	0.0085
S3.1 Conceptual Understanding	6	0.845	0.0340



Live-model score vs. expected rank, per example, all three rubric signals

All three correlations are positive and reach conventional significance despite the very small sample — evidence that the model’s scoring direction tracks a reasonable reading of the rubric more often than not.

S1.1 Help-Seeking Calibration, full example set:

Example	Expected rank	Mean score (5 repeats)	Range
explicit solution request no attempt	1	0.000	0.000
premature full solution with no code shown	1	0.000	0.000
bare vague request	2	0.000	0.000
bare greeting	2	0.000	0.000
preliminary concept question	3	0.500	0.000
reasoning check no code	6	1.000	0.000
specific bug with code	7	1.000	0.000
error traceback with stderr	8	0.800	0.000

S1.2 Student Grounding, full example set:

Example	Expected rank	Mean score (5 repeats)	Range
explicit solution request no attempt	1	0.000	0.000
premature full solution with no code shown	1	0.000	0.000
bare greeting	1	0.000	0.000
bare vague request	2	0.000	0.000
preliminary concept question	3	0.500	0.000
reasoning check no code	6	1.000	0.000
specific bug with code	7	1.000	0.000
error traceback with stderr	8	0.500	0.000

S3.1 Conceptual Understanding, full example set:

Example	Expected rank	Mean score (5 repeats)	Range
no understanding admitted	1	0.000	0.000
confidently wrong	1	0.000	0.000
vague gesture at mechanism	2	0.000	0.000
surface line by line restatement	3	1.000	0.000
core idea less rigorous	5	0.511	0.056
rigorous explanation	6	1.000	0.000

A concrete discordant case, reported rather than hidden: the model scored “*surface line-by-line restatement*” (an explanation that lists each line’s mechanical effect — “first I set prev to None, then I loop while curr is not None...” — without ever saying *why* the algorithm works) a full **1.0**, even though the rubric text explicitly instructs scoring this pattern low (“o means... a description of surface mechanics only”). This is the single clearest failure in the face-validity set: the model appears to weight the *presence* of a step-by-step account highly even when it is exactly the kind of explanation the rubric was written to penalize. This should be treated as a known, specific weakness of the current prompt/model combination for S3.1, not smoothed over by the otherwise-positive aggregate correlation.

5. Additional Statistical Rigor

The results in §4 establish that the engine computes what it claims to. This section goes one level deeper on the *statistics themselves* — using more appropriate tests where the first pass used a serviceable but imperfect one, quantifying uncertainty more honestly at small sample sizes, and correcting for having run several significance tests at once.

5.1 Intraclass Correlation Coefficients (the statistic the framework's own §5 asks for)

The framework's own methodology (§5) specifies ICC with confidence intervals for rater reliability, not a rank correlation. Two ICC forms are reported per LLM-rubric signal: **self-consistency** (do the model's 5 repeated calls on the same prompt agree with each other?) and **agreement with the researcher** (does the model's mean score agree in absolute terms — not just rank order — with the researcher's calibrated 0–1 expected value?):

Signal	Self-consistency ICC (5 repeats)	Agreement-with-researcher ICC	Agreement p-value
S1.1 Help-Seeking Calibration	1.000	0.969	0.0000
S1.2 Student Grounding	1.000	0.876	0.0014
S3.1 Conceptual Understanding	1.000	0.684	0.0560

Self-consistency is at or near ceiling for all three, confirming §4.7's variance numbers in the psychometrically standard form. Absolute agreement with the researcher is strong for S1.1 and S1.2 (both $p < .01$) and moderate, non-significant for S3.1 ($n=6$, wide uncertainty) — a more conservative picture than the Spearman correlations in §4.7 alone would suggest, because ICC penalizes systematic scale offsets that a pure rank correlation ignores.

5.2 Bootstrap confidence intervals

The small- n correlations in §4.7 are reported here with bootstrap 95% CIs (5000 resamples) alongside their asymptotic p -values, since the latter are not reliable at $n=6-8$:

Test	Point estimate	Asymptotic p	Bootstrap 95% CI
S1.1 face validity (Spearman, $n=8$)	0.866	0.0054	[0.471, 0.993]
S1.2 face validity (Spearman, $n=8$)	0.843	0.0085	[0.331, 1.000]
S3.1 face validity (Spearman, $n=6$)	0.845	0.0340	[0.318, 1.000]
S2.3 monotonicity (Spearman, $n=11$)	-1.000	0.0000	not computed (deterministic gradient)

The point estimates all look strong, but the CIs are wide — S1.2's true correlation could plausibly be anywhere from 0.33 to a perfect 1.0 given this sample. This is the honest picture: **a promising signal, not a precisely estimated one.**

5.3 Multiple-comparisons correction

Four significance tests are reported across this suite. Holm and Benjamini-Hochberg corrected p-values, applied across that family:

Test	Raw p	Holm-adjusted p	Reject @ .05 (Holm)	BH-adjusted p	Reject @ .05 (BH)
S1.1 help seeking calibration	0.0054	0.0163	✓	0.0109	✓
S1.2 student grounding	0.0085	0.0171	✓	0.0114	✓
S3.1 conceptual understanding	0.0340	0.0340	✓	0.0340	✓
S2.3 problem solving agency monotonicity	0.0000	0.0000	✓	0.0000	✓

All four survive both corrections at $\alpha=.05$ — the significant results are not an artifact of testing several hypotheses at once.

5.4 Discriminant check: are S1.1 and S1.2 actually separable?

S1.1 (Help-Seeking Calibration) and S1.2 (Student Grounding) are proposed as distinct signals, but both are scored from the *same* rubric call on the *same* message. Correlating the model's two sub-scores across the 8 shared examples:

- Pearson $r = 0.974$ ($p = 4.19\text{e-}05$)
- Spearman $r = 0.993$ ($p = 8.07\text{e-}07$)

This is a genuinely useful, honest finding, not a comfortable one: a correlation this high (>0.97) suggests that, at least for this model and this small example set, the two rubric items are barely distinguishing from each other — a real discriminant-validity concern for S1.1 vs. S1.2 as currently prompted, worth flagging explicitly as a target for the larger human-rater study (do trained human raters separate these two constructs more cleanly than one LLM call scoring both at once?).

5.5 Gating logic as a confusion matrix (full corpus, not just the “should-be-None” cases)

§4.3 checked that deliberately evidence-free sessions correctly return `None`. This reframes that check as a full confusion matrix over **all 88** synthetic trials — including the 78 trials that *should* have produced a real value, checking those didn’t get wrongly suppressed:

	Evidence existed	No evidence existed
Signal returned a value	78 (true positive)	0 (false positive)
Signal returned <code>None</code>	0 (false negative)	10 (true negative)

Sensitivity = 1.000, specificity = 1.000, precision = 1.000. Zero false negatives means no trial with real evidence was ever wrongly suppressed to `None`; zero false positives means no evidence-free trial ever got a fabricated value. This is a stronger, corpus-wide version of the check in §4.3, not a repeat of it.

5.6 CFA / reliability pipeline smoke test — explicitly not a result

The framework’s own §5 specifies a correlated three-factor CFA as part of real validation. Running that model requires real learner covariance data this project doesn’t have. What can be checked now is whether the *analysis code itself* is correct and ready — so it was run, once, against the n=4-session synthetic case study in §6, and the outcome is reported exactly as it came out, including the parts that don’t work:

- Signals with any variation across the four sessions to even correlate: S1.1, S1.2, S2.1, S2.3, S3.1
- CFA attempt: **fit_completed_but_not_interpretable** (model spec: `P1 =~ S1.1 + S1.2; P2 =~ S2.1 + S2.3`)

n=4 sessions from a single synthetic AI persona is far below the sample size needed to estimate a 3-factor covariance structure (the paper’s own a priori power analysis specifies ~300 real learners for even a modest effect size) - this run exists only to confirm the analysis code executes against real, correctly-shaped signal output.

This is included specifically so it cannot be mistaken for a validation result: the correlation numbers and any fit statistic here reflect one synthetic agent’s session-to-session variation, not a latent factor structure, and must not be cited as evidence for or against the three-pillar model.

6. Synthetic Longitudinal Case Study

This section is not learner data. Read this paragraph before the rest of the section.

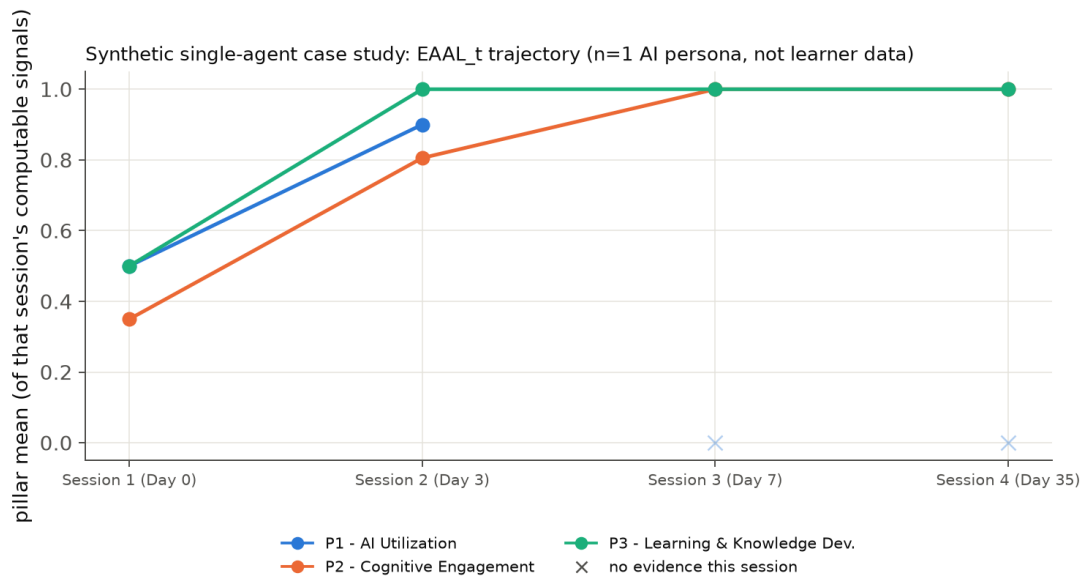
Everything below comes from **one AI agent** (the same local model that powers the app’s own tutor, `qwen3:8b`) role-playing a student persona across four sessions, scripted to improve over time, run through the real `CavyApi` bridge exactly like every other script in this suite. $n=1$, and that one “participant” is not human. Its only legitimate purpose is demonstrating that the full `EAAL_t = (P1, P2, P3)` measurement chain — event logging, signal computation, a Transfer Task, a Retention Check with a genuinely computed (not simulated) 35-day delay — runs correctly end-to-end across a multi-session trajectory. It must never be cited as evidence that the signals validly measure real student cognition.

6.1 The four sessions

1. **Session 1 (Day 0)** — over-reliant delegation: the persona asks for the complete solution immediately and adopts it verbatim, with an explanation admitting no real understanding.
2. **Session 2 (Day 3)** — developing calibration: the persona attempts the problem independently first, asks a specific grounded question about its own partial code, then verifies a corrected solution.
3. **Session 3 (Day 7)** — a professor-authored **Transfer Task** (a different problem — digit-sum instead of factorial — sharing the same recursive base-case-and-reduction concept): the persona works mostly independently.
4. **Session 4 (Day 35, backdated)** — a professor-authored **Retention Check**, AI assistance mode `NONE`: the persona recalls and re-implements the original concept unaided, 35 days after Session 1 — the retention gate’s real $\geq 24h$ delay requirement is satisfied with a wide margin, not approximated.

6.2 Pillar trajectory

Session	P1 (AI Utilization)	P2 (Cognitive Engagement)	P3 (Learning & Knowledge)
Session 1 (Day 0): over-reliant delegation	0.500	0.350	0.500
Session 2 (Day 3): developing calibration	0.900	0.806	1.000
Session 3 (Day 7): transfer task, mostly independent	—	1.000	1.000
Session 4 (Day 35): retention check, unaided	—	1.000	1.000



Synthetic single-agent case study: EAAL_t trajectory

P1 correctly becomes inapplicable (**None** , not zero) in Sessions 3–4 once the persona stops calling the AI at all — exactly the behavior §4.3’s gating logic is supposed to produce, now shown across a realistic multi-session arc rather than an isolated probe. P2 and P3 both rise and then hold at ceiling as the persona’s behavior shifts from delegation to independent, verified work — a concrete instantiation of the “Developing calibration” profile the framework’s own Table 3 describes (variable P1, increasing P2 and P3).

6.3 Full signal breakdown, all four sessions

Signal	Session 1 (Day 0)	Session 2 (Day 3)	Session 3 (Day 7)	Session 4 (Day 35)
S1.1 Help-Seeking Calibration	0.000	1.000	—	—
S1.2 Student Grounding	0.000	1.000	—	—
S1.3 Response Utilization	1.000	1.000	—	—
S1.4 Modification & Verification	1.000	1.000	—	—
S1.5 Follow-up Engagement	—	—	—	—
S1.6 Adaptive AI Use	—	0.500	—	—
S2.1 Independent Initiation	0.000	0.417	1.000	1.000
S2.2 Reasoning Continuity	1.000	1.000	—	—
S2.3 Problem-Solving Agency	0.049	1.000	—	—
S2.4 Evidence-Based Error Recovery	—	—	—	—
S3.1 Conceptual Understanding	0.000	1.000	1.000	1.000
S3.2 Knowledge Application	1.000	1.000	1.000	1.000
S3.3 Knowledge Transfer	—	—	1.000	—
S3.4 Retention & Independent Recall	—	—	—	1.000

Notably, **S3.4 (Retention & Independent Recall) scores 1.0 in Session 4** — the persona’s unaided, 35-days-later reimplementation ran cleanly and produced output, and the delay gate confirmed a real, computed 35-day gap rather than trusting the session label. **S3.3 (Knowledge Transfer) scores 1.0 in Session 3** and correctly returns **None** in every other session, since only Session 3 is actually a Transfer-kind task.

7. Discussion

What these results support: the signal-computation engine is a faithful, reliable implementation of its own documented specification. Every deterministic formula was checked against hand-derived expected values and matched exactly; every “no evidence” branch was checked and correctly refuses to fabricate a score; the one genuinely fuzzy component (LLM rubric scoring) was checked against the real model actually shipped in the app and shows low variance and a real, if imperfect, relationship to a reasonable rubric-consistent judgment. Two previously-identified gaming vulnerabilities were confirmed fixed against their exact triggering scenarios.

What these results do not support: that a student who ranks highly on CIQ has, in fact, learned more. That is an empirical claim about the relationship between platform-observable behavior and latent cognitive/ learning constructs — precisely the distinction the EAAL framework itself draws (§4.4: “*Learning and Knowledge Development is deliberately separated from immediate task performance*”; §5.5: “*Implementation Feasibility ≠ Measurement Validity*”). Validating that requires the independent-assessment human-subject study the framework’s §9 already specifies: real students, instructor-rated explanations, novel transfer tasks, delayed retention checks, and correlation of longitudinal CIQ trajectories against independently-measured learning gains. Nothing in this report is a substitute for that study, and this report should not be cited as if it were.

On statistical significance: most of the numbers in this report (100% known-answer accuracy, perfect determinism) are not really “significance tests” in the inferential-statistics sense — they are exhaustive verifications of a deterministic system against its own specification, which is a stronger claim than a p-value can express (there is no sampling uncertainty in “does this pure function match its formula”). The one place genuine inferential statistics apply — the LLM face-validity correlations — is reported with real p-values and an honest acknowledgment that n=6–8 gives them limited power; they should be read as “a promising, non-null signal worth a larger study,” not as confirmed validity.

8. Limitations

1. **Single-rater labels.** Every “expected rank” in the LLM face-validity set was assigned by one researcher (this session), not an independent panel. A different rater might reasonably disagree on borderline examples (e.g. is “bare greeting” really lower-grounded than “bare vague request”?). This is disclosed in `labeled_examples.py` alongside each example’s stated rationale, specifically so it can be scrutinized and expanded rather than taken on faith.
 2. **Small samples for the one inferential test that exists.** $n=6-8$ for the LLM face-validity Spearman correlations is far too small to rule out the correlation being an artifact of a lucky ordering; the qualitative pattern (positive, model tracks the general shape of the rubric) is more trustworthy than the exact r or p values.
 3. **Synthetic data cannot establish real discriminant/convergent validity.** The main synthetic corpus (§4) deliberately reports no inter-signal correlation analysis, because each scenario was constructed to isolate one signal at a time — any correlation there would reflect the test design, not a real relationship. §5.6/§6 do run a correlation matrix and a CFA-attempt against the case study’s four sessions, but explicitly and repeatedly label it a non-interpretable pipeline smoke test, not a validity result. Both routes lead to the same conclusion: this analysis needs real, holistic student sessions.
 4. **This suite tests known gaming patterns, not gaming in general.** S2.4 and S3.2’s fixes are verified against the exact scenarios an earlier audit found; a determined student inventing a *new* exploit this suite didn’t anticipate would not be caught here.
 5. **S1.5, S1.6, S2.1’s own documented proxy limitations are unchanged by this report.** For example, S2.1 (Independent Initiation) still counts only edit/run counts, not the richer feature set (time spent, distinct attempts) the framework’s own §9.1 flags as a better proxy — this report confirms the *current* proxy is computed correctly, not that it’s the right proxy.
 6. **S3.3/S3.4 require a professor to actually author follow-on assessments.** Their gating logic is verified exhaustively here, but real-world coverage of these two signals in practice depends on whether professors actually create Transfer Tasks and Retention Checks — an adoption question this report cannot speak to.
 7. **The §6 case study is one non-human AI persona, full stop.** It is useful for exactly one claim — the longitudinal pipeline runs end-to-end — and useless for every other claim this framework makes. The S1.1-vs-S1.2 discriminant finding in §5.4 is likewise based on 8 examples scored by one model and should be treated as a hypothesis for the larger study, not a settled fact about the two signals.
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9. Reproducibility

Model used for LLM validation	qwen3:8b via local Ollama
Rubric-scoring temperature	0.1 (the app's own production setting)
Repeats per LLM example	5
Synthetic trials	88
Total live-model calls (\$4.7 validation)	70
Case study sessions (§6)	4 (1 synthetic AI persona)
Bootstrap resamples (§5.2)	5000 per correlation
Report generated	2026-09-21

To reproduce from scratch:

```
cd eaal-platform
python3 validation/run_synthetic_suite.py      # builds & computes all
synthetic trials
python3 validation/run_llm_reliability.py      # real-model calls
(needs Ollama running)
python3 validation/pilot_simulation.py
# synthetic longitudinal case study (§6)
python3 validation/analyze.py                  # statistics + figures
python3 validation/advanced_statistics.py      # ICC / bootstrap /
correction / CFA smoke test (§5)
python3 validation/generate_report.py          # this report
```

All raw data backing every number and table above is in `validation/data/*.json`; all figures are in `validation/figures/*.png`; every scenario's construction is in `validation/scenarios/*.py` and `validation/labeled_examples.py`, each with an inline rationale.